

ALPHA ARTS AND SCIENCE COLLEGE

(AFFILIATED TO UNIVERSITY OF MADRAS)

Porur, Chennai - 600 116



A PROJECT REPORT ON

EduGenie: Google Gemini Powered Learning Assistant

Submitted as partial fulfillment of the requirements under the

NAAN MUDHALVAN (NM) SKILL DEVELOPMENT PROGRAMME

SUBMITTED BY

Student Name	NM ID
S.srinivasan	06928E0493D847B8A6043444C0A7C43B
R.Samuvel	74A264ED0D6A8CDA56A430C14871B FC3
S. saravanan	7A739303795BE958F45F365A67520F
P. Sabarinathan	C7629DE4EBC8236A9F84D0DD13C0507F

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EduGenie AI

AI-Powered Personal Learning Assistant

1. Introduction

EduGenie AI is an educational web application designed to help students learn more effectively using artificial intelligence. The system provides interactive tools for asking questions, explaining topics, generating quizzes, summarizing study material, and creating personalized learning paths.

2. Objectives

- Provide students with a simple AI-based learning assistant.
- Explain difficult topics in clear and student-friendly language.
- Generate practice quizzes automatically.
- Summarize long study materials into useful revision notes.
- Create personalized learning plans based on topic, level, and duration.
- Provide all major learning features through one web interface.

3. Technologies Used

- Python – backend programming
- Django REST API – web application/API framework
- Google Gemini API – AI content generation
- HTML – frontend structure
- CSS – frontend styling
- JavaScript – frontend interaction and API calls
- inja2 – HTML template rendering
- python-dotenv – secure environment-variable loading
- Gunicorn – development server

4. Project Structure

EDUGENIE

static

style.css

templates

index.html

explanation_module.py

learning_path.py

main.py

qna.py

quiz_module.py

requirements.txt

summary_module.py

.env

5. Main Modules

main.py – Main FastAPI application. Defines the web page and API endpoints.

qna.py – Handles student questions and generates AI answers.

explanation_module.py – Explains educational topics according to the student's level.

quiz_module.py – Generates multiple-choice quizzes with answers and explanations.

summary_module.py – Summarizes educational text for quick revision.

learning_path.py – Creates personalized study plans.

templates/index.html – Frontend interface containing the EduGenie features.

static/style.css – Controls the appearance, layout, cards, buttons, and mobile responsiveness.

requirements.txt – Contains the Python packages required to run the project.

.env – Stores the Gemini API key and model configuration securely.

6. Features

Ask EduGenie: Students can enter an educational question and receive an AI-generated answer.

Explain a Topic: Students can choose a learning level and receive a structured explanation.

Generate Quiz: Students can select a topic, number of questions, and difficulty level.

Summarize Text: Students can paste study material and receive a concise summary.

Learning Path: Students can create a study plan using a topic, level, and duration.

7. Application Workflow

1. Student opens the EduGenie web application.
2. Student selects a learning feature.
3. The frontend sends the request to the FastAPI backend.
4. The relevant Python module prepares an AI prompt.
5. The Gemini API generates the requested educational content.
6. The backend returns the result to the frontend.
7. The student views the result in the browser.

8. API Endpoints

Method	Endpoint	Purpose
GET	/	Open the EduGenie interface
GET	/health	Check application status
POST	/qa	Answer a student question
POST	/explain	Explain a topic
POST	/quiz	Generate a quiz
POST	/summarize	Summarize text
POST	/learn/recommendations	Create a learning path

9. Running the Project

1. Open the EduGenie folder in VS Code.
2. Open Terminal New Terminal.
3. Create a virtual environment using: `py -m venv .venv`
4. Activate it using: `.\.venv\Scripts\Activate.ps1`
5. Install packages using: `pip install -r requirements.txt`
6. Create a `.env` file and add the Gemini API key.
7. Start the server using: `python -m uvicorn main:app --reload --port 5000`
8. Open `http://127.0.0.1:5000/` in the browser.

10. Expected Output

The browser displays a dashboard titled "EduGenie AI – Your Personal AI Learning Assistant". It contains cards for Ask EduGenie, Explain a Topic, Generate Quiz, Summarize Text, and Learning Path. Each feature accepts student input and displays an AI-



EduGenie AI

Your AI Powered Learning Assistant



Explain a Topic

environment

Explain

Hello! I'm **EduGenie**, your AI educational assistant. Let's break down the concept of the **environment** into clear, easy-to-understand sections.

1. Simple Definition

The **environment** is everything that surrounds us. It includes all the living things (like people, plants, and animals) and non-living things (like air, water, sunlight, and soil) that interact with one another on Earth.



Learning Path

environment

Create Path

Here is a simple, step-by-step learning path on the topic of **The Environment**, designed for a college student.

Phase 1: Beginner Concepts (Foundations)

1. Understand Ecosystems and Biomes

Learn what an ecosystem is (living things interacting with non-living things). Study major Earth biomes such as forests, deserts, grasslands, and oceans.

2. Learn About Biodiversity

Understand biodiversity (the variety of life on Earth) and why it is important for stable ecosystems, food security, and human health.

3. Study Natural Cycles

Learn how nature recycles elements. Focus on the water cycle, carbon cycle, and nitrogen cycle, and see how they support life.

4. Identify Natural Resources

Differentiate between renewable resources (sun, wind, water) and non-renewable resources (coal, oil, natural gas).

Generate Quiz

environment |

Generate Quiz

Here is a 5-question environmental science quiz designed for college students:

Question 1

Which process describes the nutrient enrichment of a body of water—often due to agricultural runoff containing nitrogen and phosphorus—leading to excessive algal growth and oxygen depletion?

- A) Bioaccumulation
- B) Eutrophication
- C) Salinization
- D) Ocean Acidification

Answer: B) Eutrophication

Explanation: Eutrophication occurs when excess nutrients enter an aquatic ecosystem, causing rapid algal blooms. As these algae die and decompose, bacteria consume large amounts of dissolved oxygen in the water,

? Ask EduGenie

how it is helpful to live happy

Ask

Hello there! I am **EduGenie**, your learning buddy.

Living a happy life is not just a nice feeling—it is actually very good for your body, your mind, and your life!

Here are four main ways being happy helps you:

1. It Keeps Your Body Healthy

When you are happy, your body produces good chemicals that keep you strong.

* **Less Sickness:** Happy people get sick less often because their body fights off germs better.

* **Better Sleep:** Happy people usually sleep better at night and wake up feeling fresh.

* **Example:** Think about a day you felt really happy—you probably had lots of energy to run around and play!



Text Summarizer

The environment includes all living and non-living things around us, such as humans, plants, animals, air, water, and soil. It provides essential resources for life and maintains ecological balance. Pollution can harm the environment, so we should protect it by planting trees, reducing waste, saving water, and conserving natural resources.

Summarize

Here is a simple summary of the text:

1. Main idea

The environment gives us everything we need to live, so we must protect it from harm.

2. Important points

- * The environment includes living things (humans, plants, animals) and non-living things (air, water, soil).
- * It provides resources for life and keeps nature in balance.
- * Pollution damages the environment.
- * We can help protect the environment by planting trees, saving water, making less trash, and saving resources.

3. Key terms

- * **Environment:** Everything around us, both living and non-living.

EduGenie AI combines a simple web interface with generative AI to create a personalized learning assistant. It can support students with explanations, question answering, quizzes, summaries, and structured learning paths. The modular Python design also makes the project easy to extend with additional educational features in the future.